

4.1 角的定义及弧度(整合习题)

(4.1 Definition and Radian Measure of Angles (Integrated Exercises))

一、选择题(Multiple Choice)

1. 1010° 是第 () 象限角。

(1010° is an angle in the () quadrant.)

- A. 一(I) B. 二(II) C. 三(III) D. 四(IV)

2. 下列角中终边与 320° 相同的角是 ()。

(Which of the following angles has the same terminal side as 320° ?)

- A. 40° B. -40° C. 620° D. -620°

3. -1060° 写成 $\alpha + 2k\pi$ ($k \in \mathbb{Z}, 0 \leq \alpha < 2\pi$) 的形式是 ()。

(Express -1060° in the form $\alpha + 2k\pi$ ($k \in \mathbb{Z}, 0 \leq \alpha < 2\pi$) ().)

- A. $-\frac{\pi}{9} - 6\pi$ B. $\frac{2\pi}{9} - 6\pi$ C. $\frac{\pi}{9} - 6\pi$ D. $-\frac{2\pi}{9} - 6\pi$

二、填空题(Fill in the Blanks)

1. $45^\circ = \underline{\hspace{2cm}} \pi$; $315^\circ = \underline{\hspace{2cm}} \pi$; $-20^\circ = \underline{\hspace{2cm}} \pi$;

2. $\frac{13}{3}\pi = \underline{\hspace{2cm}}^\circ$; $\frac{3}{5}\pi = \underline{\hspace{2cm}}^\circ$

3. 把下列角写成 $360^\circ \cdot k + \alpha (k \in \mathbb{Z}, 0^\circ \leq \alpha < 360^\circ)$ 或 $2k\pi + \alpha (k \in \mathbb{Z}, 0 \leq \alpha < 2\pi)$ 的形式:

(Express the following angles in the form $360^\circ \cdot k + \alpha$ ($k \in \mathbb{Z}$, $0^\circ \leq \alpha < 360^\circ$) or $2k\pi + \alpha$ ($k \in \mathbb{Z}$, $0 \leq \alpha < 2\pi$):)

380° _____; _____;

$$\frac{11}{4}\pi = \underline{\hspace{2cm}}; \underline{\hspace{2cm}}; \circ$$

4. -1130° 在第_____象限; 1495° 在第_____象限;

$-\frac{19}{4}\pi$ 在第____象限。

(-1130° lies in the ____ quadrant; 1495° lies in the ____ quadrant; $-\frac{19}{4}\pi$ lies in the ____ quadrant.)

5. 锐角是第____象限角;钝角是第____象限角。

(An acute angle is a __ quadrant angle; an obtuse angle is a __ quadrant angle.)

三、计算题(Calculation)

1. 计算: $220^\circ = __\pi$; $20^\circ = __\pi$ 。

2. 把下列角写成 $2k\pi + \alpha (k \in \mathbb{Z}, 0 \leq \alpha < 2\pi)$ 或 $k \cdot 360^\circ + \alpha (k \in \mathbb{Z}, 0^\circ \leq \alpha < 360^\circ)$ 的形式: (Express the following angles in the form $2k\pi + \alpha (k \in \mathbb{Z}, 0 \leq \alpha < 2\pi)$ or $k \cdot 360^\circ + \alpha (k \in \mathbb{Z}, 0^\circ \leq \alpha < 360^\circ)$.)

(1) $\frac{11\pi}{4}$

(2) 1160°

3. 判断下列角的终边在第几象限: (Determine which quadrant the terminal side of each of the following angles lies in:)

(1) -70° _____ (2) 595° _____ (3) -960° _____

4.2 三角函数的定义及关系(整合习题)

(4.2 Definition and Relations of Trigonometric Functions (Integrated Exercises))

一、选择题(Multiple Choice)

1. 角 α 的终边经过点 $P(-4,3)$, 则 $\cos\alpha = (\quad)$ 。

(The terminal side of angle α passes through point $P(-4,3)$, then $\cos\alpha = (\quad)$.)

A. $-\frac{4}{3}$ B. $\frac{4}{5}$ C. $-\frac{4}{5}$ D. $\frac{3}{5}$

2. 如果 $6\sin\alpha \cdot \cos\alpha < 0$, 那么 α 在 $(\quad)$ 。

(If $6\sin\alpha \cdot \cos\alpha < 0$, then α lies in $(\quad)$.)

A. 第一、二象限(Quadrants I and II) B. 第一、三象限(Quadrants I and III)
C. 第一、四象限(Quadrants I and IV) D. 第二、四象限(Quadrants II and IV)

3. 下列运算正确的是 $(\quad)$ 。

(Which of the following operations is correct $(\quad)$?)

A. $\cos\frac{\pi}{3} = \frac{1}{2}$ B. $\sin\frac{\pi}{4} = \frac{\sqrt{3}}{2}$ C. $\tan\frac{\pi}{3} = \frac{\sqrt{2}}{2}$ D. $\sin\frac{\pi}{2} = 0$

4. 已知 $\cos\alpha = -\frac{1}{4}$, α 在第二象限, 则 $\sin\alpha = (\quad)$ 。

(Given $\cos\alpha = -\frac{1}{4}$ and α is in Quadrant II, then $\sin\alpha = (\quad)$.)

A. $\frac{\sqrt{15}}{4}$ B. $-\frac{\sqrt{15}}{4}$ C. $\frac{1}{4}$ D. $-\frac{1}{4}$

5. 已知 $\tan\alpha = -\frac{3}{4}$, α 在第二象限, 则 $\cos\alpha = (\quad)$ 。

(Given $\tan\alpha = -\frac{3}{4}$ and α is in Quadrant II, then $\cos\alpha = (\quad)$.)

A. $\frac{4}{5}$ B. $-\frac{4}{5}$ C. $\frac{3}{5}$ D. $-\frac{3}{5}$

6. 已知 $\sin\alpha = \frac{4}{5}$, $\cos\alpha = \frac{3}{5}$, 那么 $\tan\alpha = (\quad)$ 。

(Given $\sin\alpha = \frac{4}{5}$ and $\cos\alpha = \frac{3}{5}$, then $\tan\alpha = (\quad)$.)

A. $\frac{4}{3}$ B. $\frac{3}{4}$ C. $\frac{4}{5}$ D. $\frac{3}{5}$

7. 已知 $\sin\alpha = -\frac{3}{5}$, α 在第三象限, 则 $\cos\alpha = ()$ 。

(Given $\sin\alpha = -\frac{3}{5}$ and α is in Quadrant III, then $\cos\alpha = ()$.)

A. $-\frac{3}{5}$ B. $\frac{4}{5}$ C. $-\frac{4}{5}$ D. $\frac{4}{3}$

二、填空题(Fill in the Blanks)

1. 判断下列三角函数值的正负:

(Determine the sign (positive/negative) of the following trigonometric values:)

(1) $\cos(-440^\circ)$ _____; (2) $\tan\left(-\frac{19\pi}{8}\right)$ _____; (3) $\sin 158^\circ$ _____;

(4) $\sin 105^\circ$ _____; (5) $\cot 210^\circ$ _____; (6) $\cos\left(-\frac{1}{8}\pi\right)$ _____。

2. 计算: $\sin(-80^\circ) + \cos 10^\circ + \tan 50^\circ =$ _____。

3. 计算: $\cos\frac{\pi}{3} + 3\tan\frac{\pi}{6} - \sin\frac{\pi}{6} =$ _____。

4. 已知 $\tan\alpha = -\sqrt{2}$, α 在第二象限, 则 $\cos\alpha =$ _____。

*(Given α and α is in Quadrant II, then .)

5. 已知 $\tan\alpha = 3$, 则 $\frac{3\sin\alpha - 2\cos\alpha}{2\sin\alpha + 3\cos\alpha} =$ _____。

(Given α , then .)

三、计算题(Calculation)

1. 判断下列三角函数值的正负:

(Determine the sign (positive/negative) of the following trigonometric values:)

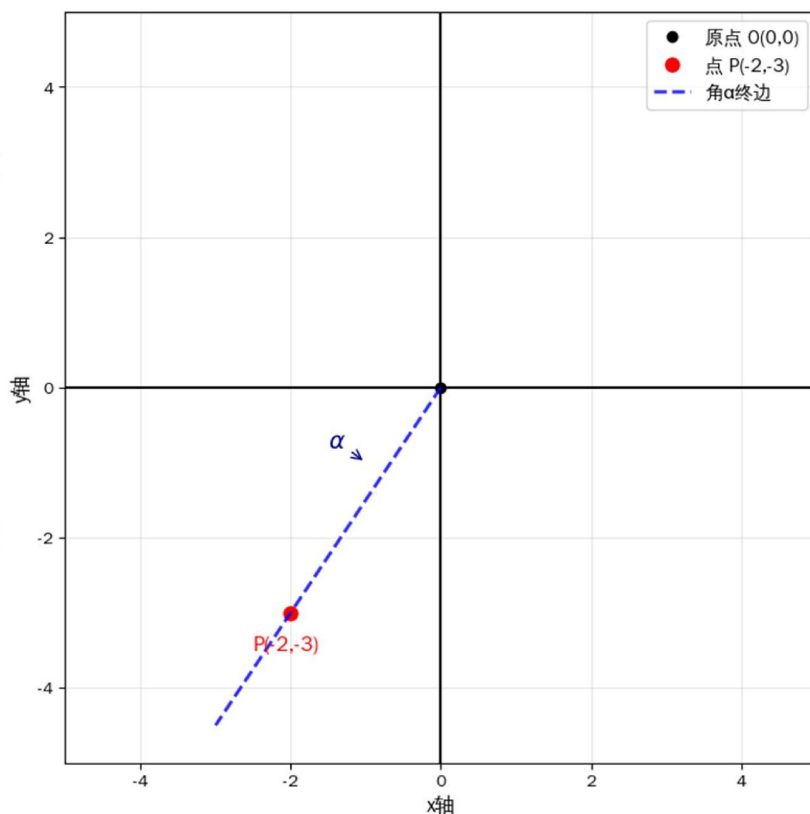
(1) $\sin 220^\circ$ _____ (2) $\cos 98^\circ$ _____

(3) $\tan\frac{4\pi}{5}$ _____ (4) $\cot\frac{23\pi}{6}$ _____

2. 已知角 α 的终边经过点 $P(-2, -3)$, 求角 α 的六个三角函数值。

(The terminal side of angle α passes through point $P(-2, -3)$, find the six trigonometric values of α .)

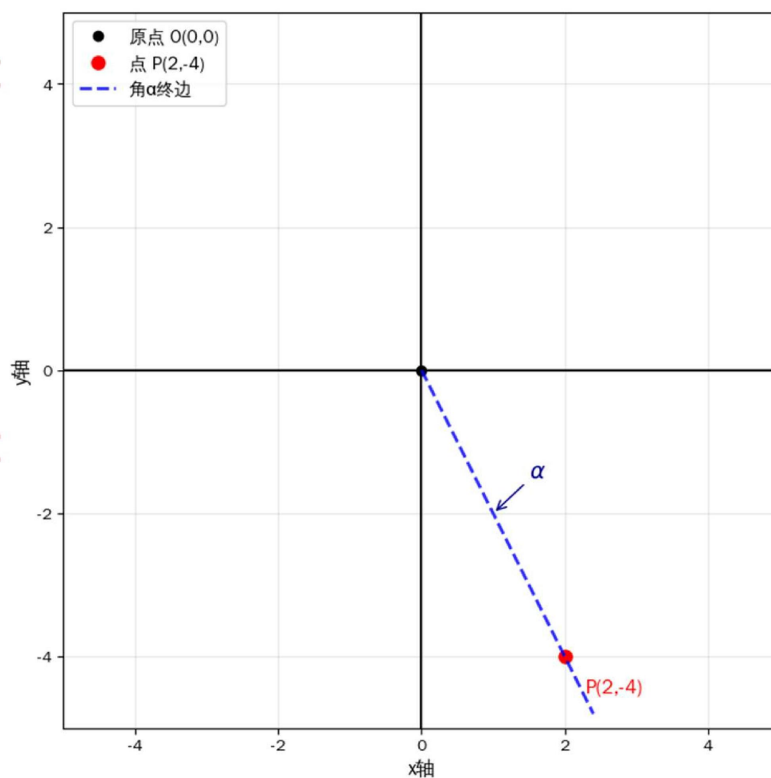
点 $P(-2, -3)$ 及角 α 终边示意图



3. 已知角 α 的终边经过点 $P(2, -4)$, 求角 α 的六个三角函数值。

(The terminal side of angle α passes through point $P(2, -4)$, find the six trigonometric values of α .)

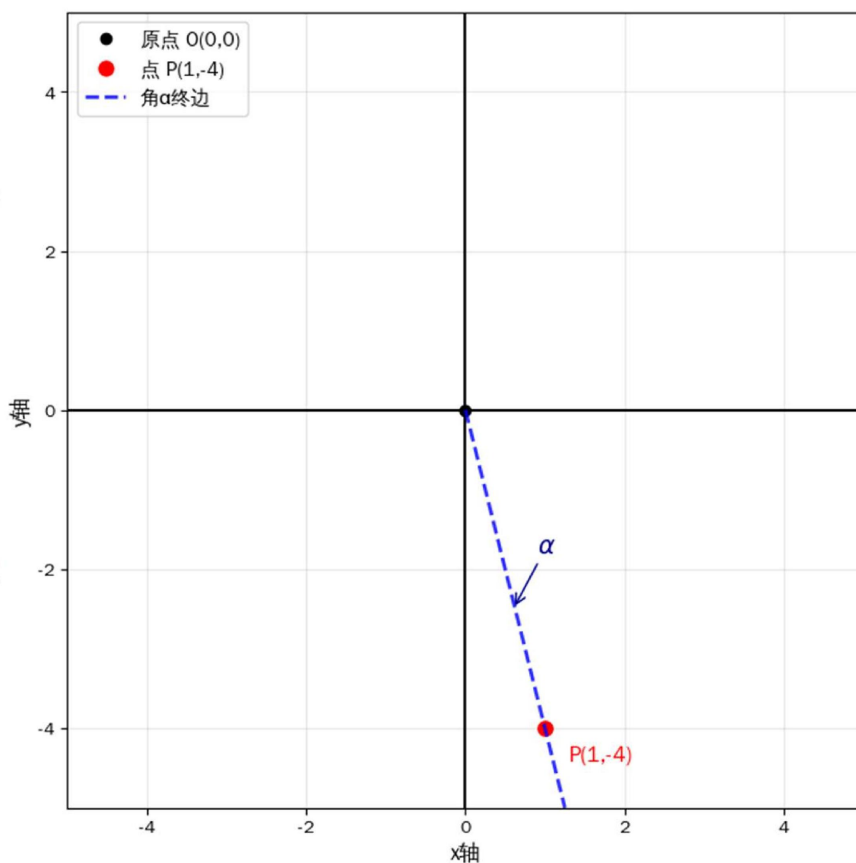
点 $P(2, -4)$ 及角 α 终边示意图



4. 已知角 α 的终边经过点 $P(1, -4)$, 求角 α 的六个三角函数值。

(The terminal side of angle α passes through point $P(1, -4)$, find the six trigonometric values of α .)

点 $P(1, -4)$ 及角 α 终边示意图



5. 已知 $\sin\alpha = -\frac{3}{5}$, α 是第三象限角, 求 $\cos\alpha$ 、 $\tan\alpha$ 、 $\cot\alpha$ 的值。

(Given $\sin\alpha = -\frac{3}{5}$ and α is a Quadrant III angle, find the values of $\cos\alpha$, $\tan\alpha$, and $\cot\alpha$.)

6. 已知 $\sin\alpha = \frac{\sqrt{3}}{2}$, 角 α 在第二象限, 求 $\cos\alpha$ 、 $\tan\alpha$ 、 $\cot\alpha$ 的值。

(Given $\sin\alpha = \frac{\sqrt{3}}{2}$ and α is in Quadrant II, find the values of $\cos\alpha$, $\tan\alpha$, and $\cot\alpha$.)

7. 已知 $\tan\alpha = \frac{4}{5}$, $\alpha \in (\pi, \frac{3}{2}\pi)$, 求 $\cos\alpha$ 、 $\sin\alpha$ 的值。

(Given $\tan\alpha = \frac{4}{5}$ and $\alpha \in (\pi, \frac{3}{2}\pi)$, find the values of $\cos\alpha$ and $\sin\alpha$.)

4.3 三角函数的诱导公式(整合习题)

(4.3 Induction Formulas of Trigonometric Functions (Integrated Exercises))

一、选择题(Multiple Choice)

1. $\sin \frac{5\pi}{6} = ()$ 。

A. $-\frac{1}{2}$ B. $\frac{1}{2}$ C. $\frac{\sqrt{3}}{2}$ D. $-\frac{\sqrt{3}}{2}$

2. $\tan\left(-\frac{11\pi}{6}\right) = ()$ 。

A. $-\sqrt{3}$ B. $-\frac{\sqrt{3}}{3}$ C. $\frac{\sqrt{3}}{3}$ D. $\sqrt{3}$

3. $\sin \alpha = ()$ 。

A. $\sin(\pi - \alpha)$ B. $\sin(\pi + \alpha)$ C. $\sin(-\alpha)$ D. $\sin\left(\frac{\pi}{2} + \alpha\right)$

二、填空题(Fill in the Blanks)

1. 求下列三角函数值：(Find the values of the following trigonometric functions:)

$\tan 600^\circ = \underline{\hspace{2cm}}; \sin\left(-\frac{17\pi}{4}\right) = \underline{\hspace{2cm}}; \cos(-945^\circ) = \underline{\hspace{2cm}};$

$\cos 210^\circ = \underline{\hspace{2cm}}; \tan 150^\circ = \underline{\hspace{2cm}}; \sin\left(-\frac{26\pi}{3}\right) = \underline{\hspace{2cm}};$

$\cot 780^\circ = \underline{\hspace{2cm}}; \tan \frac{19\pi}{4} = \underline{\hspace{2cm}}; \cos \frac{10\pi}{3} = \underline{\hspace{2cm}}。$

2. 如果 $\sin \alpha = \frac{1}{2}$, α 在第二象限, 那么 $\alpha = \underline{\hspace{2cm}}。$

(If $\sin \alpha = \frac{1}{2}$ and α is in Quadrant II, then.)

三、计算题(Calculation)

1. 求下列三角函数值：

(Find the values of the following trigonometric functions:)

(1) $\sin 235^\circ$ _____ (2) $\cos \frac{13\pi}{3}$ _____ (3) $\cos 400^\circ$ _____

(4) $\sin \frac{11\pi}{4}$ _____ (5) $\tan\left(-\frac{13\pi}{6}\right)$ _____

2. 化简求值: $\frac{\cos(3\pi+\alpha)}{\sin(-\alpha)} \cdot \tan(-\alpha - 2\pi)$ 。

(Simplify and evaluate: $\frac{\cos(3\pi+\alpha)}{\sin(-\alpha)} \cdot \tan(-\alpha - 2\pi)$.)

3. 化简求值: $\frac{\cos(3\pi+\alpha)}{\sin(-\alpha)} \cdot \tan(\pi - \alpha)$ 。

(Simplify and evaluate: $\frac{\cos(3\pi+\alpha)}{\sin(-\alpha)} \cdot \tan(\pi - \alpha)$.)

4.4 正弦函数和余弦函数的图像和性质(整合习题)

(4.4 Graphs and Properties of Sine and Cosine Functions (Integrated Exercises))

一、选择题(Multiple Choice)

1. 下列是 $y = \sin x$ 的增区间的是 ()。

(Which of the following is an increasing interval of $y = \sin x$ ()?)

A. $[-\frac{\pi}{2}, \frac{\pi}{2}]$ B. $[\frac{\pi}{2}, \frac{3\pi}{2}]$ C. $[0, \pi]$ D. $[\pi, 2\pi]$

2. $y = 3\cos(\frac{1}{4}x)$ ($x \in \mathbb{R}$) 是 ()。

($y = 3\cos(\frac{1}{4}x)$ ($x \in \mathbb{R}$) is a () function.)

A. 奇函数(Odd) B. 偶函数(Even)

C. 既是奇函数又是偶函数(Both odd and even)

D. 非奇非偶函数(Neither odd nor even)

3. $y = 3\sin(\frac{1}{4}x + \frac{\pi}{6})$ 的最小正周期是 ()。

(The minimum positive period of $y = 3\sin(\frac{1}{4}x + \frac{\pi}{6})$ is ().)

A. 2π B. 4π C. 6π D. 8π

4. $y = 4\cos 3x$ ($x \in \mathbb{R}$) 的最小正周期是 ()。

(The minimum positive period of $y = 4\cos 3x$ ($x \in \mathbb{R}$) is ().)

A. 2π B. $\frac{2\pi}{3}$ C. 4π D. $\frac{\pi}{3}$

5. 函数 $y = -4\sin x$ 在 $[0, 2\pi]$ 上的单调性是 ()。

(The monotonicity of the function $y = -4\sin x$ on $[0, 2\pi]$ is ().)

A. 在 $[0, \pi]$ 上是增函数, 在 $[\pi, 2\pi]$ 上是减函数

(Increasing on $[0, \pi]$ and decreasing on $[\pi, 2\pi]$)

B. 在 $[\frac{\pi}{2}, \frac{3\pi}{2}]$ 上是增函数, 在 $[0, \frac{\pi}{2}] \cup [\frac{3\pi}{2}, 2\pi]$ 上是减函数

(Increasing on $\left[\frac{\pi}{2}, \frac{3\pi}{2}\right]$ and decreasing on $\left[0, \frac{\pi}{2}\right] \cup \left[\frac{3\pi}{2}, 2\pi\right]$)

C. 在 $\left[0, \frac{\pi}{2}\right]$ 上是增函数, 在 $\left[\frac{\pi}{2}, \frac{3\pi}{2}\right]$ 上是减函数

(Increasing on $\left[0, \frac{\pi}{2}\right]$ and decreasing on $\left[\frac{\pi}{2}, \frac{3\pi}{2}\right]$)

D. 在 $\left[0, \frac{\pi}{2}\right] \cup \left[\frac{3\pi}{2}, 2\pi\right]$ 上是增函数, 在 $\left[\frac{\pi}{2}, \frac{3\pi}{2}\right]$ 上是减函数

(Increasing on $\left[0, \frac{\pi}{2}\right] \cup \left[\frac{3\pi}{2}, 2\pi\right]$ and decreasing on $\left[\frac{\pi}{2}, \frac{3\pi}{2}\right]$)

二、填空题(Fill in the Blanks)

1. $y = \sin x$ 的最小正周期是____;最大值和最小值是____,____。

(The minimum positive period of $y = \sin x$ is ____; its maximum and minimum values are ____.)

2. $y = 4\sin x$ 的最小正周期是____;最大值和最小值是____,____。

(The minimum positive period of $y = 4\sin x$ is ____; its maximum and minimum values are ____.)

3. $y = 4\sin \frac{1}{3}x$ 的最小正周期是____;最大值和最小值是____,____。

(The minimum positive period of $y = 4\sin \frac{1}{3}x$ is ____; its maximum and minimum values are ____.)

4. $y = 4\sin\left(\frac{1}{3}x + \frac{\pi}{7}\right)$ 的最小正周期是____;

最大值和最小值是____,____。

(The minimum positive period of $y = 4\sin\left(\frac{1}{3}x + \frac{\pi}{7}\right)$ is ____; its maximum and minimum values are ____.)

5. $y = 4\sin \frac{1}{3}x + 2$ 的最小正周期是____;

最大值和最小值是____,____。

(The minimum positive period of $y = 4\sin \frac{1}{3}x + 2$ is ____; its maximum and

minimum values are ____.)

6. $y = -3\cos x$ 的最小正周期是____;最大值和最小值是____,____。

(The minimum positive period of $y = -3\cos x$ is ____; its maximum and minimum values are ____.)

7. $y = -3\cos 4x$ 的最小正周期是____;最大值和最小值是____,____。

(The minimum positive period of $y = -3\cos 4x$ is ____; its maximum and minimum values are ____.)

8. $y = -3\cos\left(4x - \frac{\pi}{7}\right)$ 的最小正周期是____;

最大值和最小值是____,____。

(The minimum positive period of $y = -3\cos\left(4x - \frac{\pi}{7}\right)$ is ____; its maximum and minimum values are ____.)

9. $y = -3\cos 4x + 4$ 的最小正周期是____;

最大值和最小值是____,____。

(The minimum positive period of $y = -3\cos 4x + 4$ is ____; its maximum and minimum values are ____.)

10. $y = 3\cos x$ 的奇偶性是____;在 $(-\pi, \pi)$ 上的减区间是____。

(The parity of $y = 3\cos x$ is ____; its decreasing interval on $(-\pi, \pi)$ is ____.)

11. $y = \sin x + 2$ 的奇偶性是____;在 $(0, \pi)$ 上的增区间是____。

(The parity of $y = \sin x + 2$ is ____; its increasing interval on $(0, \pi)$ is ____.)

12. $y = 5\cos x - 4$ 的奇偶性是____;

$y = 5\sin x - 4$ 的奇偶性是____。

(The parity of $y = 5\cos x - 4$ is ____, the parity of $y = 5\sin x - 4$ is ____.)

13. $y = \cos\left(x - \frac{\pi}{8}\right)$ 的奇偶性是_____;

$y = \sin\left(x - \frac{\pi}{8}\right)$ 的奇偶性是_____。

(The parity of $y = \cos\left(x - \frac{\pi}{8}\right)$ is _; the parity of $y = \sin\left(x - \frac{\pi}{8}\right)$ is _.)

三、计算题(Calculation)

1. 求下列函数的最小正周期:

(Find the minimum positive period of the following functions:)

(1) $y = 6\sin x \ (x \in \mathbb{R})$

(2) $y = \cos 4x \ (x \in \mathbb{R})$

(3) $y = \sin\left(\frac{3}{4}x + \frac{\pi}{5}\right) \ (x \in \mathbb{R})$

2. 写出下列函数的最大值和最小值:

(Write the maximum and minimum values of the following functions:)

(1) $y = \cos 3x$

(2) $y = \sin 3x + 2$

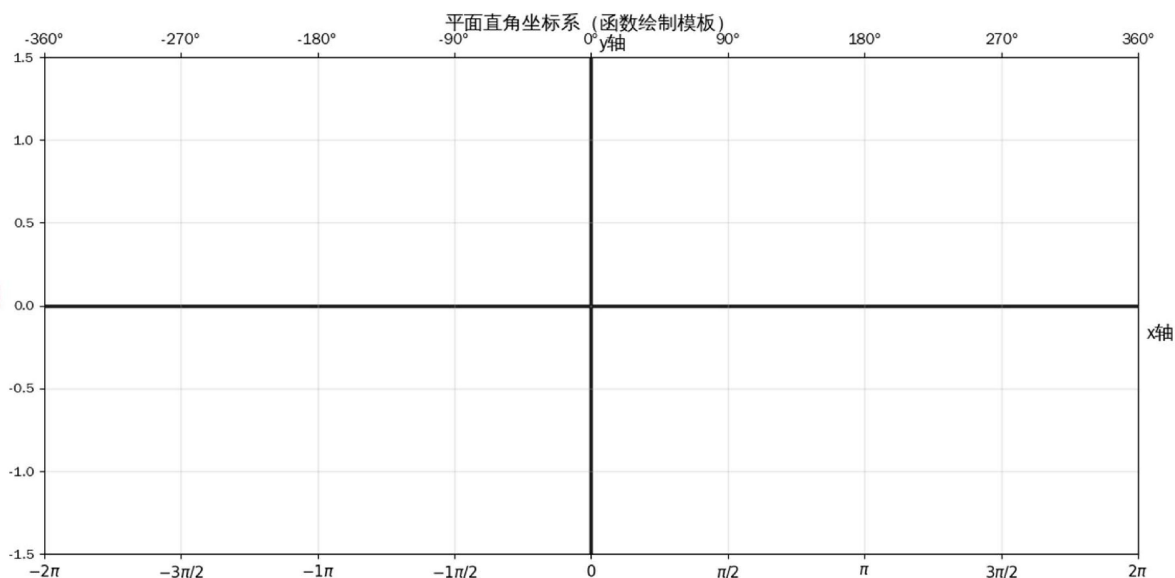
(3) $y = 4\sin x$

四、画图题(Graphing)

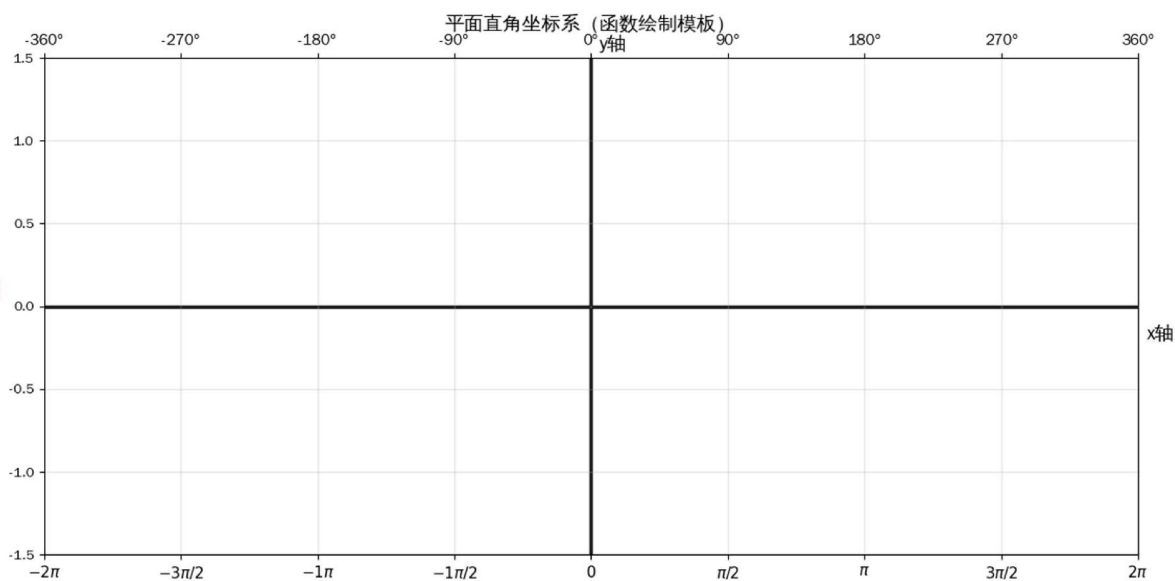
画出下列函数的图像：

(Sketch the graphs of the following functions:)

(1) $y = -2\sin x$;



(2) $y = 4\cos x$



4.5 正切函数和余切函数的图像和性质(整合习题)

(4.5 Graphs and Properties of Tangent and Cotangent Functions (Integrated Exercises))

一、选择题(Multiple Choice)

1. $y = 4\tan\frac{x}{5}$ 的最小正周期是 ()。

(The minimum positive period of $y = 4\tan\frac{x}{5}$ is ().)

A. 2π B. 3π C. 4π D. 5π

2. $y = \tan\left(x + \frac{\pi}{5}\right)$ 的定义域是 ()。

(The domain of $y = \tan\left(x + \frac{\pi}{5}\right)$ is ().)

A. $\mathbb{R}$ B. $\{x \mid x \neq k\pi + \frac{\pi}{2}, k \in \mathbb{Z}\}$ C. $\{x \mid x \neq k\pi + \frac{3\pi}{10}, k \in \mathbb{Z}\}$ D. $\{x \mid x \neq \frac{\pi}{5}\}$

3. $y = \tan\left(3x + \frac{\pi}{4}\right)$ 的奇偶性是 ()。

(The parity of $y = \tan\left(3x + \frac{\pi}{4}\right)$ is ().)

A. 奇函数(Odd) B. 偶函数(Even) C. 非奇非偶函数(Neither odd nor even) D.

无法确定(Cannot be determined)

4. $y = \cot 4x$ 的奇偶性是 ()。

(The parity of $y = \cot 4x$ is ().)

A. 奇函数(Odd) B. 偶函数(Even) C. 非奇非偶函数(Neither odd nor even) D.

无法确定(Cannot be determined)

二、填空题(Fill in the Blanks)

1. $y = \tan 3x$ 的最小正周期是_____;

定义域是_____。

(The minimum positive period of $y = \tan 3x$ is _____; its domain is _____.)

2. $y = -5\tan\frac{x}{3}$ 的最小正周期是_____;

定义域是_____。

(The minimum positive period of $y = -6\tan\frac{x}{4}$ is ____; its domain is ____.)

3. $y = \frac{1}{3}\tan\left(x + \frac{\pi}{7}\right)$ 的最小正周期是____;

定义域是_____。

(The minimum positive period of $y = \frac{1}{3}\tan\left(x + \frac{\pi}{7}\right)$ is ____; its domain is ____.)

4. $y = \cot\frac{x}{3}$ 的最小正周期是____;

定义域是_____。

(The minimum positive period of $y = \cot\frac{x}{3}$ is ____; its domain is ____.)

5. $y = \cot 3x + 2$ 的最小正周期是____;

定义域是_____。

(The minimum positive period of $y = \cot 3x + 2$ is ____; its domain is ____.)

三、计算题(Calculation)

求函数 $y = \tan\left(\frac{1}{3}x + \frac{\pi}{4}\right)$ 的定义域和最小正周期。

(Find the domain and minimum positive period of the function $y = \tan\left(\frac{1}{3}x + \frac{\pi}{4}\right)$.)

自测题(整合习题)

(Self-Test (Integrated Exercises))

一、选择题(Multiple Choice)

1. 函数 $y = -\sin 6x$ ($x \in \mathbb{R}$) 的最小正周期是 ()。

(The minimum positive period of the function $y = -\sin 6x$ ($x \in \mathbb{R}$) is ().)

A. 2π B. π C. $\frac{\pi}{3}$ D. $\frac{2\pi}{3}$

2. $y = 3\cot \frac{x}{4}$ 的最小正周期是 ()。

(The minimum positive period of $y = 3\cot \frac{x}{4}$ is ().)

A. 4π B. 3π C. $\frac{1}{4}\pi$ D. 2π

3. $y = 9\tan \frac{x}{3}$ 的最小正周期是 ()。

(The minimum positive period of $y = 9\tan \frac{x}{3}$ is ().)

A. 9π B. 6π C. 3π D. π

4. 函数 $y = \cos x + 6$ 是 ()。

(The function $y = \cos x + 6$ is a () function.)

A. 奇函数(Odd) B. 偶函数(Even)

C. 既是奇函数又是偶函数(Both odd and even)

D. 非奇非偶函数(Neither odd nor even)

5. 函数 $y = \cos\left(x + \frac{\pi}{7}\right)$ 是 ()。

(The function $y = \cos\left(x + \frac{\pi}{7}\right)$ is a () function.)

A. 奇函数(Odd) B. 偶函数(Even)

C. 既是奇函数又是偶函数(Both odd and even)

D. 非奇非偶函数(Neither odd nor even)

6. 函数 $y = -3\sin x + 2$ 是 ()。

(The function $y = -3\sin x + 2$ is a () function.)

- A. 奇函数(Odd) B. 偶函数(Even)
C. 既是奇函数又是偶函数(Both odd and even)
D. 非奇非偶函数(Neither odd nor even)

7. 函数 $y = 4\sin\left(x + \frac{\pi}{6}\right)$ 的奇偶性是 ()。

(The parity of the function $y = 4\sin\left(x + \frac{\pi}{6}\right)$ is ().)

- A. 奇函数(Odd) B. 偶函数(Even)
C. 非奇非偶函数(Neither odd nor even)
D. 减函数(Decreasing function)

8. $y = 3\sin x$ 的减区间是 ()。

(The decreasing interval of $y = 3\sin x$ is ().)

- A. $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ B. $\left[\frac{\pi}{2}, \frac{3\pi}{2}\right]$ C. $[0, \pi]$ D. $[\pi, 2\pi]$

9. $\sin \frac{10\pi}{3} =$ ()。

- A. $\frac{1}{2}$ B. $\frac{\sqrt{3}}{2}$ C. $-\frac{1}{2}$ D. $-\frac{\sqrt{3}}{2}$

二、填空题(Fill in the Blanks)

1. $-130^\circ =$ _____ π ; $145^\circ =$ _____ π ;

$\frac{11}{4}\pi =$ _____ $^\circ$; $\frac{3\pi}{4} =$ _____ $^\circ$ 。

角 1230° 在第_____象限。(The angle 1230° lies in the _____ quadrant.)

2. 若 α 是钝角, 那么 $\frac{\alpha}{3}$ 在第_____象限。

(If α is an obtuse angle, then $\frac{\alpha}{3}$ lies in the _____ quadrant.)

3. $\sin 30^\circ =$ _____; $\tan 30^\circ =$ _____; $\tan 60^\circ =$ _____; $\cos 420^\circ =$ _____。

4. 角 α 的终边经过点 $(-2, -3)$, 则 $\cot\alpha =$ _____。

(The terminal side of angle α passes through point $(-2, -3)$, then $\cot\alpha =$.)

5. 求值: $\sin\frac{8\pi}{6} =$ ____; $\cos 600^\circ =$ ____; $\tan(-225^\circ) =$ _____.

6. $y = \frac{1}{3}\sin\left(3x + \frac{\pi}{4}\right)$ 的最小正周期是_____;

最大值是____;奇偶性是_____。

(The minimum positive period of $y = \frac{1}{3}\sin\left(3x + \frac{\pi}{4}\right)$ is ____; its maximum value is ____; its parity is ____.)

7. $y = 4\cos x$, $x \in (-\pi, \pi)$ 的减区间是_____。

(The decreasing interval of $y = 4\cos x$ for $x \in (-\pi, \pi)$ is ____.)

8. $y = 4\tan\frac{1}{3}x$ 的定义域是____;最小正周期是_____。

(The domain of $y = 4\tan\frac{1}{3}x$ is ____; its minimum positive period is ____.)

9. $\tan 300^\circ + \cot 405^\circ =$ _____。

三、计算题(Calculation)

1. 已知 $\tan\alpha = 4$, 求 $\frac{5\sin\alpha - 3\cos\alpha}{6\cos\alpha + 4\sin\alpha}$ 的值。(Given $\tan\alpha = 4$, find the value of $\frac{5\sin\alpha - 3\cos\alpha}{6\cos\alpha + 4\sin\alpha}$.)

2. 已知 $\cos\alpha = \frac{1}{5}$, $\frac{3}{2}\pi < \alpha < 2\pi$, 求 $\sin\alpha$ 和 $\tan\alpha$ 的值。(Given $\cos\alpha = \frac{1}{5}$ and $\frac{3}{2}\pi < \alpha < 2\pi$, find the values of $\sin\alpha$ and $\tan\alpha$.)